MATH 201 Your name: ______________________

Learning target AD2, version 2

Suppose that for some function $p(x)$, you know the following information:

- $p(-2) = 5$,
- $p'(-2) = 1.5$,
- $p''(x) < 0$ for x-values close to -2.

1. Explain and demonstrate how to find the linearization $L(x)$ of $p(x)$ at $x = -2$.

2. Explain and demonstrate how to estimate the value of $p(-2.03)$ using this linearization.

3. Is your estimate of $p(-2.03)$ greater than or less than the actual value? How do you know?

4. Sketch a possible graph of $p(x)$ and its linearization $L(x)$ nearby $x = -2$ to illustrate your findings. Label important points in your sketch with their coordinates.

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Learning target AD2, version 3

The funny thing about square roots is that most of the time, they are *irrational* numbers, which means that they can't be written as fractions, and thus their decimal expansions go on forever and never repeat. For instance, $\sqrt{67} = 8.185352771872449969953703724733929458880486815498039963066715\text{2}0$ That is clearly not a particularly useful answer to the question "what's $\sqrt{67}$?" We might in particular prefer to come up with a *fraction* that is a reasonably good approximation.

(a) The number $\sqrt{67}$ is an output y-value of the function $f(x) = \sqrt{x}$. What's the corresponding input x-value?

(b) Think of an x-value that's close to the input value you said in part (a), but for which it is *easy* to figure out the square root.

$f(\_\_\_\_) = \sqrt{} = \_\_\_\_$

(c) Compute $f'(x)$ and plug in your easy x-value. Leave your answer as a fraction.

$f'(\_\_\_\_) = \_\_\_\_$

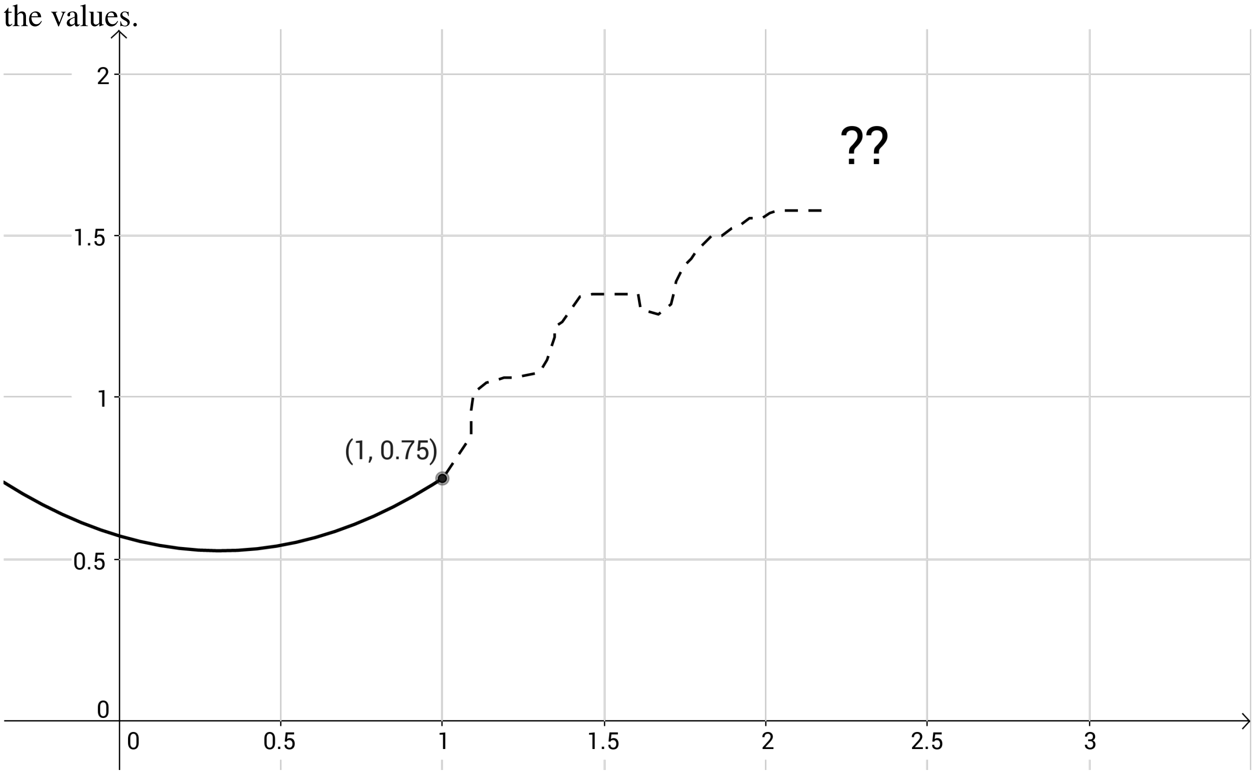
(d) Write down the equation for the tangent line to $f(x)$ at the easy x-value.

$L(x) =$

(e) Lastly but not leastly, plug in the hard x-value into the equation for the tangent line. Leave your answer as a fraction.

Learning target AD2, version 4

Part of the graph of this function $f(x)$ has been erased. We'll use linear approximation to recover the values.



(a) Draw the tangent line to this function at $x = 1$.

(b) Suppose we know that $f'(1) = 0.65$. Use point-slope form to write down an equation for $L(x)$, the tangent line to this function at $x = 1$.

(c) Use your equation for $L(x)$ to estimate $f(1.4)$.

(d) Do you think it is more likely that this is an overestimate or an underestimate? How come?